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5 September 1975
SUPERSEDING
MIL-E-1/1072E(NAVY)
30 July 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 22CP7A AND 22CP25A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection and focus, aluminized

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rgl	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0	450	18,000	-125	Note 1	10,000
Minimum:	5.7	-125	---	---	---	---	
TEST CONDITIONS:	6.3	Adjust	300	15,000	---	---	

GENERAL:

Qualification - Required

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
(F)	Qualification inspection						
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	---	15	μAdc
5223	Modulation	-	Ib = 200 μAdc	ΔEc1	---	35	Vdc
5216	Cathode illumination	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all Control grid to all Grid No. 2 to all	Ck Cg1 Cg2	--- --- ---	8 10 10	pF pF pF
1141	Pressure (implosion)	-	35 lb/psi	---	---	---	---
5111	Vibration	-		Width	---	2.5	mm
1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1						
---	Radius of faceplate	-		---	140	---	Inches
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5206	Gas ratio	-		Gr	---	0.25	---
5101	Neck and bulb alinement (magnetic types)	7		---	---	0.400	Inch
5101	Face tilt	8		---	---	0.375	Inch
5101	Neck straightness	-		---	---	---	---
5106	Bulb, screen, and faceplate quality	6		---	---	---	---
5231	Spot position (magnetic deflection)	-		---	---	25	mm
5236	Zero-bias anode current (magnetic deflection)	4		---	---	---	---
5241	Grid cutoff voltage	-		Ec1	-33	-77	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5216	Strav light emission (conventional types)	-	Eb = 18.000 Vdc; Ec2 = 450 Vdc	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5221	Screen (P7)	2		---	---	---	---
---	Screen (P25)	5		cBf D0.1(10) D1(10) D1(10)-D10(10)	600 400 325 50	--- --- --- 80	cB cB cB cB
5226	Line width A	3	Ib = 200 μAdc	Width	---	1.25	mm
5226	Line width C	3	Ib = 200 μAdc	Width	---	2.0	mm
5246	Focusing ampere turns	-	Ib = 200 μAdc; D = 3.25	AT	600	720	---
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
1111	Base pin solder depth	-		---	---	---	---
(F) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(F) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C: Eb = 14,000 Vdc; Ec2 = 450 Vdc; Ib = 200 μAdc	t	500	---	hours
---	Life-test end points:						
5226	Line width A	-	Ib = 200 μAdc	Width	---	1.35	mm
5226	Line width C	-	Ib = 200 μAdc	Width	---	2.2	mm
5223	Modulation	-	Ib = 200 μAdc	ΔEc1	---	39	Vdc

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.

2. The screen characteristics shall be measured with constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 4,000 V minimum and beam current 60 μA maximum. The screen characteristics shall comply with the following minimum limits: cB5=400 cB:G5:1-4.

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NOTES: -Continued

3. Use focus coil JEDEC No. 109 or No. 122 and 70 degree deflection coil JEDEC No. 118.

4. For P25 phosphor, apply full raster to prevent screen burning.

5. The phosphorescent light intensity observed(t)seconds after completion of the nth raster to the tube is Dt(N). cBf shall be the fluorescent light output under continuous raster exciation. The screen characteristics shall be measured under the following conditions.

   (a) Raster size (focused beam) 50 cm^2 with tube defocused to a spot approximately 0.25 cm in diameter.
   (b) Beam current. The tube operated at a current to produce 0.05 watt at the specified test voltage.
   (c) Duration of raster. 1/60 second.
   (d) Distance of raster from axis of calibrated 931VA multiplier tube. 30 cm.
   (e) Red or white light. Sufficient to de-excite the phosphor before build-up measurements so that the results are within 3 cB of those that would have been obtained after total de-excitation.

6. The maximum diameters of "dead" and "color" spots shall be 0.060 inch.

7. The tube shall be supported vertically on its face and rotated by a suitable adjustable turntable mechanism which will permit the neck to run true to the center of rotation of the turntable. A dial indicator perpendicular to the cone surface and 3/8 inch above the back of the faceplate metal lip shall be used to determine the amount of neck and bulb misalinement. The variation shall be not greater than the specified limit.

8. With the tube supported as specified in note 7, the face tilt shall not exceed 0.375 inch.

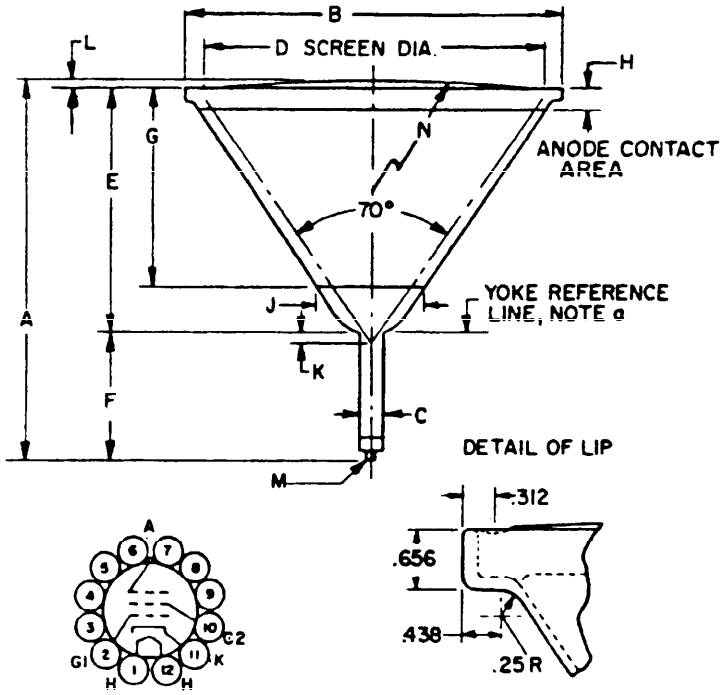
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
M	Base: B5-57	
Quality conformance inspection, part 2		
A	---	22.063 (560.40)
B	21.625 (549.28)	21.875 (555.63)
C	1.375 (34.93)	1.500 (38.10)
D	20.000 (508.00)	---
E	13.625 (346.08)	14.000 (355.60)
F	7.312 (185.72)	7.688 (195.28)
H	0.875 (22.23)	---
J	---	6.375 (161.93)
L	0.188 (4.78)	0.438 (11.13)
Reference dimensions (see note b)		
G	11.375 (288.93)	
K	1.15 (29.21)	
N	165.00 (4191.00) Rad.	

NOTES:
a. Reference line determined by position where reference line gage #110 will rest on glass funnel.
b. These dimensions are for reference only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 22CP7A and 22CP25A.

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